ActInf Textbook Group ~ Cohort 6 ~ Session 19 (Chapter 8, Part 1) 8/26/2024

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okay we're back in cohort 6 for chapter 8 first meeting so enter however you wna do it nice sound was good um yes so welcome today we're talking about chapter eight in the 2022 active inference textbook by friston par and kulo and uh chapter8 is following chapter 7 where we discuss discreet State space models um as we know the second half of the textbook is largely centered around actually designing your own active inference models as opposed to uh primarily verbal descriptions of the theory in the first half so in chapter eight which complements chapter seven we turn to continuous State space models which the authors state are well suited for modeling physical uations and pinging on sensory receptors and for the continuous motion of the affectors EG muscles we used to change the world around us uh we're presented with model examples for the dynamical systems involved in motor control as well as the concept of generalized synchrony and finally constructing Hybrid models for combining discrete and continuous variables at differing levels uh section 8.2 moves on to movement control uh we're introduces some notation for the stochastic equations involved in continuous time models which we'll notice are a little bit different from those in chapter 7 um and then we also differentiate the the symbols between those of the generative model and those of the those of the generative process which is also a reminder that oftentimes we produce models whose um construction is actually rather similar equation wise to those the generative process an example for motor control and reflex arcs uh viewed as descending proprioceptive signals is given which describes how State predictions can be pulled towards these signals like point detractors we also see how action is involved in uh true State changes in uh the generative process to bring outcomes closer to what the model expects and further how more complex forms of these equations are used in more typical use cases uh as with discreete time models continuous time ative models are used to draw inferences about the causes of Sensations B basing and brain and then uh some simple nomenclatures so in continuous uh State space models usually it's X is the new standin to refer to States Y is now for observations or incoming sensory data and U there there are different ways of of doing it but we also have like for example you can use an Omega to to represent um another aspect like different kinds of fluctuations it actually starts so one of the equations starts to look like a like a linear uh equation model so like a yal g ofx plus Omega um so kind of like a yals MX plus b or or an error term for Omega and there there's quite a bit in this chapter so I'll just give a bit of a summation of the rest 8.3 were introduced to different kinds of dynamical system which better rep kind of illustrate the idea of how attractors come to play in continuous models or or or shown the uh lka voltera Dynamics which follow from uh Predator prey Dynamics researched in say ecology or zoology with an example of three different populations plants herbivores and carnivores uh showing a kind of oscillatory pattern between their populations the idea being that carnivores eat the herbivores uh who eat the plants and so they're all kind of dependent upon one another and they start to show these kind of kind of uh non equilibrium like steady state kind of dynamics that that arise over time and it's as if there's a these kind of attracting points that they they arrive around but never actually land and settle there um we also have luren systems talking about things like criticality um then we'll we'll skip ahead a little bit Section 8 4 a very significant Concept in active inference is generalized synchrony they use a bird song example for modeling uh to to illustrate generalized synchrony so the bird song example for modeling communication and multi-agent inference problems uh they're uh the song birds uh take turns singing so it's a kind of turn taking problem and it's as if they're singing from the same him sheet two birds with similar models learn each other model parameters over time they start to synchronize with one another over time so their internal States and state predictions come to resemble one another reflecting A Primitive kind of theory of mind the Dynamics are modeled by a higher level whose State predictions of all according to a slow lurin system impact the parameters of a faster luren system at the lower level again with action here the birds are singing to minimize prediction errors arising when no song is heard as the Lauren system evolve modeling chaos the bird synchronization brings about what the authors call an identical synchronization of chaos um this also involves sensory attenuation another very significant concept the birds alternate between two uh between attention to sensory data and sensory attenuation to carry out actions this back and forth with two agents becomes a kind of conversation between them with the generally linear dynamic in their predictions right rising in a lower dimensional space so generalized synchrony can be applied to a range of diverse phenomena from synchronization between internal and external states such as the agent and its environment to communication social behavior and multi-agent settings like like the song bird example where where agents internal states start to uh sort of mirror those of each other then section 8.5 kind of uh concludes with discussing hybrid models which very well could have deserved its own chapter um but the you know the idea is you could have models that say are hierarchical where one layer perhaps a lower level uh as in figure 8.6 as Daniel was brought up um the lower layer could for example take in continuous data and try and model that and send up whatever kinds of predictions it makes in a different kind of form to a higher level which then makes some kind of you know discreet State space inference about that continuous data and make some kind of inference about it which then is set back down to the lower level um to affect its its its decision making and actions so um I think that's a good enough summary to stop there um and now I will open the floor any questions or points of conversation whenever it was introducing uh having a smooth space instead of a discret space like the last chapter said it kind of introduced it just like oh yeah this is what you use whenever the other one's impractical this is a little bit empirical but like have you guys found that you want to use like discreet more than anything as it kind of implies in this chapter or do you guys find that like you just pick the one that seems to be right for the data and then you run with it is there a preference or is there a norm that we use good question um I mean for from if if it's in the form of like a like a personal question like personal experience uh I I've largely just work with discrete State spaces uh I'm coming from kind of a particular like social sciences agent based modeling sort of background where things are presumed to occur in discret time steps where you know ideally your data effectively is just like you know discret logs in a spreadsheet or something like that um and so like most of the time I sort of default to thinking about uh discrete State spaces that said though um I mean I've worked with neuroimaging data I'd love to make some kind of synthetic like hybrid model that actually pulls in uh Neuro Imaging data and I would absolutely switch to a continuous State space space setting for for that right basically just I would probably focus on what kind of data it is that that your agent is receiving in the format and then take that approach as opposed to getting too creative from the start um yeah yeah another practical aspect is that pmdp only has discrete time setting so to use that at this point kind of bound there um and also separate out the state spaces from the time handling so here in the in this chapter they kind of bring both the continuous time handling and the continuous dates space like they go really well together but you also could have like a continuous variable like just between zero and one any number in the discrete time however then it's like if you think about that it's like the it's like the Matrix math that we looked at in chapter 7 doesn't apply where so if we just had four locations but location could have been a continuous variable um so I think the the fullest slash uh most technologically enabled would be to go with the data type if you're collecting from an analog sensor then like EEG or a continuous reading or the continuous location of something however even continuous measurements at some point on digital Hardware are getting sliced up and so then it's like okay how much do you want a coar grain um that influences a lot of the handling it's just simpler with discrete State space and discrete time okay cool thank you that kind of reminds me back in college we would uh talk about scraping electrons off of ccds and actually extracting that information and how it all is digital at that level gives me nightmares uh I also wanted to ask could you talk a little bit more about the uh Lorent part after the uh volara thing I didn't quite understand that as well as I understood the uh volara explanation and that is the rather I almost feel like just despite how Central the bird song example is used in this chapter I actually think it's it's the Loren system that that shows some much more complexity um the kind of complexity involved in in some of these kinds of models but um I I'll I'll um yeah go for go for it no if you if you had something sorry I'm just like yeah so okay so the the Loren is kind of a classic chaotic system it's kind of the um grandparents chaotic system and that was Loren studying the weather and saying wait even closely related trajectories diverge that's the leop exponent all that how it's being used in the chapter is to give a bu stability to turn taking so it's like here if you if all you knew was the x value you'd find okay I'm staying between I'm staying negative and then it's like whoop now there was a phase chance into being positive so that like sort of by stable nature of the luren attractor system is being used to be a a a chaotic um switch on turn taking in Birds um it's it's also kind of relevant like the birds are not sharing new information with each other so it's not exactly like turn taking in say a normal person conversation it's more like they're just wanting and preferring that The Song Remains the Same that the song's played out and then it's like as soon as it's being listened to and then when the listening um is like wait there's a missing note then the other bird steps in um that that's yeah go for okay no I'm sorry I was just goingon to say that that's what they now it makes sense that they were talking about the synchronization of chaos that really clicks for me now yes this is oh the since like 2015 when I think the fith and friston bird song comes out it's been used a lot um this 32.3 or 32 series one of the rare ones with a DOT three um this paper goes into more detail let's just look at a few of the key pieces here so here here's um sampling from a loren attractor so you have you can have a continuous smooth differentiable chaotic system meaning sensitivity to to small inputs highly Optive exponents and then you could sample probabilistically from that so you're getting sampling that's like kind of speckling approximating that um smooth curve um what is done with the smooth curve so is separate out on this landscape of Attraction where's the gradience from higher to lower density and then what is the solenoidal flow that goes on an ISO Contour so then um this is another key aspect of the continuous time in discrete time to look any time steps into the future you need to do these explicit modelbased rollouts like actually considering future time steps and all the combinatoric explosions that accompany that and could you huh could you please remind me what an ISO Contour is okay ISO Contour is like when you're hiking a line around the hill at the same elevation okay so it's just orthagonal to the the the gradient of steepest elevation is like putting a ruler on the that the steepest possible you're on a place on the hill steepest possible gradient and then you have orthagonal to that you have around the the map okay the Taylor series expansion is when you can take you the first order approximation is just the value second is the slope the red line third so as you take these higher and higher approximations you can approximate arbitrary functions going out um here they couple two chaotic systems they have them um and then they like in that state space that's described by the movement of them they show that you can apply that hel hols decomposition and say like where what are the what are the circulatory flows and then what are the gradient flows um this this is the um this is what the system looks like so this is the two birds so to speak here's their interface and so system a and system B and then these kinds of plots are are verifying like yeah some States like the third and the sixth their correlation drops to zero whereas other states remain correlated [Music] um so this the whole reason why they have Loren is just show you can put any smooth curve in the continuous it doesn't have to be just like a sinusoid um or anything like that you can have any kind of smooth differentiable curve and sample probabilistically from it so you could generate Point data from an underlying curve and you could take sampled Point data and then regress essentially against and get to a smooth differentiable setting and which one of those directions is useful just depends on what kind of data input you have okay cool that reminds me a lot of very classical machine learning whenever you consider your data points to be you know just points on a manifold and you're trying to learn the shape of that manifold that's okay that's kind of familiar to me yes and so then then sampling based approaches are trying to make a smooth differentiable landscape from the Speckles because with a smooth landscape you can do like gradient descent but you couldn't do gradient descent on just a cloud of points yes absolutely well not well yeah can jump from point to point um and so that's what is getting played with I mean on a bigger scale in the second half of the book but then explicitly in the last part of chapter 8 when they come together in the Hybrid models is like you and and then as Andrew highlighted like temperature in the world may be a continuous variable but you could model it with integers or you could model it with um .1 like one significant figure of of decimal Precision so then those are all modeling choices um and that's the kind of infinite branching Paths of how you model something okay cool thank you it's it's also in in the communication style models so kind of bird song um EP epistemic communities like the kind of Twitter hashtag version on Andrew social science simulation what becomes really interesting is like what rails are put in and then what is actually the semantics of what is communicated I'm not gonna lie that's oh I'm sorry yeah go Ah that's exactly what I was thinking about whenever you said like how you divide up temperature like specifically how you divide it up seems like it would directly go into how you are going to interpret the results so for the bird songs and stuff like that like just jumping into what frequencies you're naturally attuned to focusing on what axioms are naturally put into the system they are going to represent what you are going to more easily represent that was a poorly worded sentence but it's going to lead you to learn certain things easier which seems like it would help with the uh synchronization of chaos specifically yeah what will be like forever beyond that model's Horizon like if you're only doing temperature by the tens that but perceiving it by the tens like one significant figure then there won't be it it won't be able to capture at the ones place whereas if you had a model that was at the decimal point maybe it could kind of do something at a at or like what what kind of communication amongst agents is being set up like if you just say if you have put everything in place so it's like zero means go left and one means go right well then they're just going to be sending ones and zeros and using that to coordinate their movements but then how would you actually make a a kind of packet that convey move left but without putting the rabbit in the Hat so it's unfair so that they're just kind of hinting each other what they already know versus are you trying to explain the sort of um puristic signaling amongst agents that already have a shared semantics or on the other extreme end are you trying to model the emergence of shared communication semantics from some kind of a simpler substrate or like the birds like they already have the him sheet so they're keeping up the duty of the song being played and then you can show really interesting phenomena like in in the papers where they'll show okay bird one will transfer its parameters to two then you remove one you add three and then two can transfer to three so you can say there's a propagation of the fine tuning in this kind of genetic context where it knows the song but then that model couldn't be used to address how does the song change within or between Generations this is a little we might have discussed it the last time we discussed this this chapter in a previous cohort um so like page 164 um still withs example there's a figure 8.5 that uh represent like the synchronization and communication process and on the left we have the the generative model and the attractor system yeah then on the right synchronization manifold so so the the upper graph is supposed to represent like there's it's a little bit morea I don't want to use the word chaotic but you know it's a little less coordinated um but then after learning there is oranization where it's as if you know on the Y AIS that's the second bird um xaxis is the bird so there's this kind of secreation such that you know it looks like AAR you know positive correlation but what's interesting to Simply that the values um the value range on the X and Y axis um are much larger in the second graph and you know I would need to know much more about parameter settings Etc but I I was trying to figure out like if the birds are like further expanding their exploration of the state space as they continue to learn right because they if I don't know why that would expand outward um and I I might just be missing something very simply but it's just it's kind of interesting that it's such a smaller State space that they're awkwardly uh exploring before learning and then after learning it's broader yeah also in both cases it goes further to the right like here it never you know it never goes below 50 I mean never goes below -20 but then it it way overshoots um there might be something like a conservation of like in IA so it's like if you had no synchronization you'd have a circle I mean the path would be traced within a circle and then um if you have total synchronization it's almost like that's kind of like making this ellipse that might project out further but what are these really about well they're second level expectations why is the sensory data that's the actual note that's played into the environment level one I believe is the vocal cord of the bird that's like the warbling of the vocal chord control I don't know if it's a two note song I don't know if there's any auditory representation of these songs that would be pretty fun too level two then would probably represent like who's supposed to be singing and maybe the zero point is the switch over so then then this represents a an extreme belief like that it's plus 60 like 60 to one odds or e to the 60th or something that the second bird is singing but yeah these these Big Blocks kind of get thrown out and it's like what are they do that for one kind of the Classic this gets to the predictive coating question so I guess the first layer is the neural control of warbling and then the second layer is higher vocal Center and then it's kind of interesting like you know is the audio um is there like a mirror neuron thing where you're responding to your own generation of speech I mean there has to be because you kind of confirm that you're speaking as you expect but then this was going to be my question for you Andrew was why do you think there's sensory attenuation during action oh why not just keep High attention on the sensory inputs while you're acting sure my my sense of that from what I've understood because that's one of the most seemingly counterintuitive Concepts uh that I found in in active inference at least kind of like in a you know the way it's presented in the textbook but um the way I kind of imagine the logic of it in a more colloquial sense it's like if I if I am absolutely certain that my arm is exactly in the position that it's in and I maintain that sort of belief uh perpetually it's going to be very hard to change my mind that it is or could be anywhere else right there's a kind of like like by sort of by attenuating that incoming sensory evidence that it's exactly where it is right now um that allows me to kind of like put on pause so to speak the the feeding of that sensory information into reconfirming a belief I have that is there such that now I can actually move it what I what I need to happen is to believe that it's actually where I want it to be rather than where it currently is such that I can actually realize that sort of expectation um I don't know if any of that tracks but that's kind of you know in a simple sense like kind of how I've been understanding since re attenuation yeah I mean I remember we had many fun conversations about getting out of the chair and about disorders of initiating and terminating movement so it's kind of like discrepancy is resolved by changing your mind or changing the world so updating belief States or updating action States if we had complete total rigid confidence that and this also relates to a a a like a key multiple role for the expectations about sensory input they're both trying to be veridical like representing sensory inputs otherwise it's just it's just off kilter and and deceptive so on one hand the expectations have to be um at least tracking real world changes like we're experiencing the visual generated expectations hence the color and the sharpness and the periphery all that so from a first past like sensemaking perspective you want that visual expectation to be accurate but then if you had total Precision on what that were you wouldn't have the openness of the flexibility to take in New Or different information but that's just on the sense making now we get into the action so then if you had because there's also this role for expectations in pragmatic value because they're also the preferences about sensory inputs so if we're getting if we're getting confirmatory propri receptive input that the arms in position one then there's there's there's no discrepancy to even resolve we we expect it where it is we prefer it where it is so the best the most veritical and the highest pragmatic policy is to do nothing so then there has to be this like transient alleviation of the Precision to even enable the possibility that that um counterfactual about where the arm could be and the propriate reception that a companies that like for that to even be plausible the Precision has to be widened to include it so making things significantly less precise on on reading your incoming sensory data um that said not completely gone I almost imagine like a sort of intermittent like a almost like a blinking like is like U you know as opposed to completely continuous intake of of the propri receptive signal of where your arm is it's like kind of almost have to like Loosely check and make sure it's going in the direction you want it to not to imply that any of this is necessarily like a conscious process but but it's as if you need to kind of intermittently check that it's going the way you want to and and is in the position you want it to be in um else like I could imagine that getting a little Haywire if the entire process were paused to the entire duration of the movement but I might be off base here with how you're describing it but I see what I I do see this is the ni lead end of talking about decision yeah I mean it's so it's so well integrated with fixed eye gaze which is what the example that comes up again again at the end of the chapter just briefly fixed eye gaze High precision and then it's like you're driving on a windy road it's like pull off the Precision make the move but you don't see a blurry set of you don't see the blur it's like you're just not paying attention while the Cade is happening and then as soon as it's fixed again then the precisions re-engage and it looks clear and then shorts of memory kind of like just irons out the gap with your expectations I suppose yeah yeah yeah it's such a I don't know it's such a almost like a fun concept to like rra your head around um but it but it does make I mean it makes sense to me as far as the the other aspects of like the theoretical aspects of the active inference go it's like if you just yeah if you're fully fixed on a particular belief that you have about say the position your arm or where your eyes are Etc the ideas I mean you would be self-evidence you would not apt to change like the location that you're looking at uh right so it it's coherent despite the kind of seeming like you know what is that how does it work when first presented with it um and it does track to the way that I understand it and please let me know if you guys like agree or disagree because I might be misunderstanding is whenever I read that I kind of got the impression that like inside of us there are multiple systems or at least this is the way I imagine it there's multiple systems there's like the proceptive system there's like my eyes telling my head where my arm is and there's a whole bunch of things so every time I take an action there's going to be all of systems which are engaged and these systems want to have a lot of harmony with each other so there's going to be constant checks every time there's going to be an action we want to use this attenuation to make sure okay are all the gears shifting as they should is this transmission moving in Step so all of these systems are going to align themselves with every action so there's going to be some model in my head of the world that I see that is going to be built up by the things that I touch and feel especially through Pro reception through sensory stuff uh through up my eyes like my retinas actually take in we I think we went two chapters ago about how the eyes get like such small amounts of information and we actually build up the world around that and a few other things which are going to be a system of things and then the connection of the systems are going to build up this world model that I have in my head and that world model that I have in my head is the thing that we're using the free energy principle with that's the thing that we're minimizing a surprise with so does that kind of read that like you want to constantly have this attenuation to have these checks these systems are in stability in harmony yeah I mean the free energy minimization process applies within a modality and in the composition of modalities so you're right that for a real organism there's also these cross calibration checks um that that might do a lot of the work where in a single modality it doesn't need to be it doesn't need to be as resilient within a single modality because you can like double check you could look at something and then you could like move it to Double Or you could see what it appears to be and then touch it to get confirmatory evidence from a different and then if something is is showing up coherently across modalities it's EXT it's like multiple confirmatory lines of evidence absolutely yeah and this actually leads into another question I had which is not so related to chapter 8 but it's in chapter eight so I'm going to bring it up it's the question of what do you guys think about like having multiple ideas on the same thing kind of the superposition of ideas and just kind of picking out just wisping out the one that makes the most sense in this chapter it talks about the accurate belief that I am not moving in order to uh predict the sensory consequences of movement uh blah blah blah but I find that really interesting because it just says the accurate belief that I am not moving and that could mean a few things that depends on the frame of reference which I'm putting it that depends on the context for the actual place that I want to apply that free energy because like I'm not using it right now but I'm standing on a treadmill so I could be moving but at the same time relative to my desk I'm not moving but then again relative to the sun I'm absolutely moving but then whenever you think about all these other things there's going to be this kind of like which place do I want to put my sensory input in my sensory input is not just my sensory input it's the sensory input with the context of the environment I'm building it around where the environment I'm building it around right now is this funny little idea of I am not moving with respect to what do you guys have any thoughts on that is is that as deep as it goes is that just like I read a sentence too hard or [Laughter] this is it precisely the kind of uh you know the the kind of conundrum that that I was trying to express when I first started speaking about sensory attenuation you know it's uh it's so I keep saying seemingly counterintuitive because at this point there's something intuitive my brain that's telling me I think I get it now um but really I mean it's tough with um I mean to to be on a treadmill the difficulty with that that example is that like if you're just standing on you know a treadmill let's maybe go with an escalator uh like if you're just standing there like you're not moving like you're moving and that's accurate in a in a kind of you know external you're moving in the world's you know XYZ coordinate 3D World sense but you're not taking action you know so if if if your mark on blanket is is you you're not taking the actions to move your body and at least in the way that we're saying that and so it's I mean it's not going to be I don't know if this necessarily plays into that however if you're you know standing in place and you want to move right then like the idea of sensory attenuation is that you would have to sort of downtune or widen the Precision uh or or or otherwise uh regarding your sensory observations of standing in place at least to some you know degree oritt or however you know how it's your choice how you want to actually model that and try to get at that but however it is the ideaas you have to kind of put on pause this idea that you're just standing in place lest you just kind of reconfirm the belief that you're just standing in place you'll even if you like want to move uh that that's not going to sit well with like the rest of your like so-called beliefs as as an agent right the rest of your beliefs telling you well both of my legs are neither of them are moving and like in terms of all my senses I appear to not be moving um self- evidencing like you're G to stay there um you know even if you consciously have some belief that you want to move I mean I I think it'd be I I think it's interesting to look at that from a kind of you know psychiatric standpoint as well or looking at you know kind of phantom pain or or things like that um but uh so so you kind of I mean I think I think the way that Daniel put it earlier too with the sad I think that was really good actually like you're not actually noticing the blur necessarily whenever you move your eyes right it's kind of like like you know we predictive coding sort of think of it is like the higher the higher order is just kind of like descending and put like expectations like kind of fills in the The Gap whenever you move your eyes around but you're not actually focusing on that FL you're really downtuning kind of that sensory information you'd otherwise be taking in because you're not focused on oh my eyes are exactly in you know position 2.21 or 2.1 by 3.1 now they're in 2.0 by 3.0 like you're not necessarily traing every incremental bit that you move right you're just I'm looking here I want to look here um everything in between is just like a kind of skip and your your mind Fs in the blank so it's not too terribly jarring your everyday experience right also surprise minimization the treadmill separates which you know not necessarily an evolutionarily um prevalent stimulus separates out two senses of I am there's I am making movement which could be proprioceptive change so rate of proceptive change is I am making movement then there's I am moving distance which would relate like you said a lot to the reference frame and then those two might have it's like if you were in a dream and you felt like you were running but you felt like you were moving slow that could be like a gear differential between the appropriate receptive that you felt like you were like working hard then the visual that you weren't and then that would be interpreted as like a Salient differential yes absolutely and also I I think dreams are just deeply fascinating in the realm of uh active inference that's just amazing to me the dreams will be super fun to explore and like on the lucid dreaming like I like can they control and reduce their agency level like can they submerge or can they can they control their attention or could they you know what aspects of the generative model are being are being modified and what are the bounds of the Lucid agency what are they reducing or increasing their Precision on and then what then are there different like subtypes where where somebody might be able to remember more but they can't have agency over wanting to remember more or something like that I don't know oh that's that's a really really cool idea how much of the because you are the environment in that case there's no the separation between you know the self and the environment you're in a dream it's just it's all you so that is such a fascinating angle to go down yeah but at the same time the way you're processing that it's as if you're you're a mar on blanket inside of a Marco blanket as supp you know you're you're still distinct from you uh the world that your mind's creating I almost wonder if there's a if a dream is just some kind of like residue of of offline structure learning right like there's just there's some kind of mashing around of different you know model parameters or however you want to say and then suddenly you have this visual scene where you get to like move around in their affordances and at some point you might just notice that you have more control than what you think you do I don't know fun conversation had of run around dreams always the hybrid model at the end that was kind of most um most recently unpacked in the uh active inference does not contradict folk psychology a nice careful title not that it that it supports or or verif IES any kind of but just that it that it isn't the wrong link I'll I'll fix that later but just that it's that it um doesn't contradict and then the belief desire intention folks psychological model um and so just like can if all things can be modeled in active inference then that would include these other kinds of experience situations and other things so then like then again that kind of reflects well what what is it that allows us to describe anything are there things outside of our modeling scope or are there just situations that are just empirically very challenging to assess like you couldn't get a verbal self-report from somebody in a lucid dream or just something like that like there's situations where you couldn't get a data point but you could model where you could get a data point I'm personally of the mindset that our tools can lead us to places way outside of how creative we can be because whenever we're like building these things we're you know we have a very serious like we have our math hats on and I think of physics whenever they found like black holes and stuff like that and the big bang and whenever they looked at it they said that's completely ridiculous there's no way that's true and I think that with a lot of tools we see very similar things to where we didn't well it led to very unintuitive results yeah yeah U want to you know I think it new to apply active inference in certain areas that I'd like to you know apply it to but uh like a followup point on that is that you know so like a tend to like for centuries models that really simplify the idea of what is human behavior or something like utility or profit maximizers you know and and the kind of outcomes model that using computational usually what hens being very identical to one another it's kind of a zero some game Cut Pro to cut and because people are so identical they don't have their own preferences They Don't Really trade things with one another it's just a very flat simple environment and uh suddenly once you once you include things like like uh you know allow for more complexity suddenly you you end up having agents who trade with one another in ways that leads to Market volum ility and and crashes and Bubbles and all of these other kinds of things so what's interesting to me is like on the one hand I do agree with you I think that these kinds of tools lead to all kinds of like crazy new discoveries that I mean that's one of the greatest as the combination of like epistemic and pragmatic value there is like doing research to make new discoveries and at the same time it's like sometimes I think these tools will actually help us understand already exist existing phenomena but now we actually know how to recreate them and now better understand what their initial causes are too so um just coming at it from kind of the other angle as well no I completely agree feeling like you get to see something from a new angle using a new tool is actually one of the most satisfying things I think you can experience and now cognitive science is moving that it's not just we're taking a picture of like our elbow that we couldn't have seen it's like oh that's what my elbow looks like from that angle if it's like that's what my mind looks like from that angle or that's a view of a mind that hasn't existed then then that is just so strange loopy and trippy absolutely yeah I'm trying to remember there's some famous quote from uh I don't know if it was Richard finan or who but uh some quote like if if I can't create it and I don't understand it it was it's like a kind of wor by a model if I cannot create it I don't understand it so trying to understand things chapter seven that's where we're at we I wonder where I got that from we understand creating the discrete time models I think it's fair to say at this time we understand creating the discreet time models more not not like a some fundamental aspect of them just that we've had a lot more experience with pmdp and the discreet time maths are a little clearer but then the continuous time especially with RX and fur it'll be very cool to kind of flip in and out and make both and Nest them and and just really start to see like how how and where they can be can we could just do a continuous in a discret time or state space for like every single situation and just kind of have like two models in parallel then just recombine them as needed yeah yeah absolutely I I I imagine like uh you know I I was thinking about um how to model like for example agents who do have access to some kind of you know uh re some kind of continuously measured resource and like what's important is to be actually actually be able to measure if something is like higher or or lower like actually having that kind of like order ordinal scale or otherwise because that isn't that kind of the issue with trying to take some continuous value and then bending it such that becomes discrete choices is that you also have have to somehow include the logic that the range of 0 to 10 is necessarily less than the range of 11 to 20 right like if you if you have a model that doesn't quite get that then it's going to make associate like it might make very oddl looking associations between what it means to be in 0 to 10 versus 11 to 20 right so um anyway maybe conversation for another well next time we Explore More eight maybe we'll look at and possibly add some questions eight getting you know eighth inning after the seventh inning stretch um some some have left the stadium um but but then we we head back into the data oriented nine and the closing 10 so cool thank you fellas see you next time wonderful talk to you guys thanks soon peace bye take care